CASH CONVERSION CYCLE

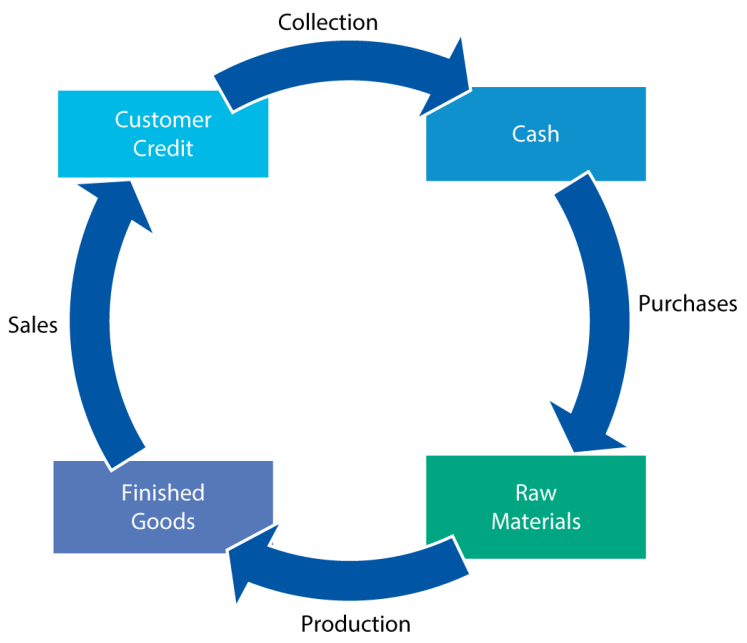
Learning Outcome

- explain the cash conversion cycle and compare issuers' cash conversion cycles

A company's business operations are usually composed of several sequential steps. For a company that makes and sells physical goods, its operations include acquiring materials, producing inventory, selling products to customers, and collecting cash. These activities are known as the issuer's **operating cycle** and occur once or many times over a year, as illustrated in Exhibit 1.

Exhibit 1:

Operating Cycle



These activities result in cash outflows and inflows that usually do *not* occur at the same time as the activity. For example, materials are purchased and received by a firm but may not be paid for in cash until weeks or months later. Goods are sold to customers, but cash may not be received until weeks or months later. Finally, inventory may take time to produce or be ready for sale, such as in the spirits industry, where an aged product like whiskey can take years before it is ready to be sold to customers.

Future cash inflows within the operating cycle are recorded as short-term assets on issuers' balance sheets, while future cash outflows within the operating cycle are recorded as short-term liabilities. Issuers' financial statements often use the terms and associated definitions in Exhibit 2 for different types of future cash inflows and outflows.

Exhibit 2:

Selected Short-Term Assets and Liabilities

Accounts receivable	Amounts to be collected from customers for products or services sold
Inventory	Cost of products produced or purchased for sale
Short-Term Liability	**Meaning**
Accounts payable	Amounts owed to suppliers for products or services received

These short-term assets and liabilities are recognized based on the performance or occurrence of an activity and are derecognized when the cash inflow or outflow occurs, as illustrated in Exhibit 3.

Exhibit 3:

Selected Short-Term Assets and Liabilities Recognition Criteria

Short-Term Asset	Recognized When . . .	Derecognized When . . .
Accounts receivable	Product or service is sold to customer on credit	Cash is received from customer
Inventory	Issuer takes ownership of materials, goods, supplies, etc.	Product is sold to customer
Short-Term Liability	**Recognized When . . .**	**Derecognized When . . .**
Accounts payable	Product or service is received, and issuer defers payment to supplier	Cash is paid to supplier

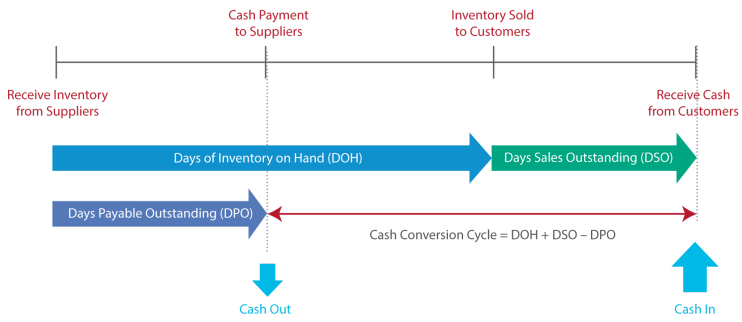
We can use the average duration of each of these short-term accounts to construct a timeline for a company's operating cycle. The amounts of time that accounts payable, inventory, and accounts receivable are outstanding on the balance sheet are known, respectively, as **days payable outstanding (DPO)**, **days of inventory on hand (DOH)**, and **days sales outstanding (DSO)**. The calculations of these amounts, known as **activity ratios**, will be discussed in detail later in the curriculum. An important quantity from this timeline for analysts is the **cash conversion cycle**: the amount of time between an issuer's paying its suppliers and receiving cash from customers (i.e., the time between derecognition of accounts payable and derecognition of accounts receivable). Formally, the cash conversion cycle, in days, is expressed in Equation 1 as:

$$\text{Cash conversion cycle} = \text{Days of inventory on hand} + \text{Days sales outstanding} - \text{D payable outstanding} \quad (1)$$

Exhibit 4 depicts the cash conversion cycle on a timeline within the broader operating cycle.

Exhibit 4:

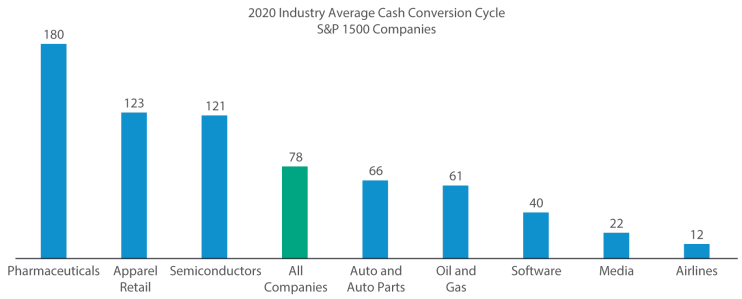
The Cash Conversion Cycle



The cash conversion cycle is the number of days it takes a company to convert an inventory investment into cash receipts from customers. Therefore, the longer the cash conversion cycle, the longer a company needs financing to pay its bills, such as payroll, because it has not yet received cash from customers. In 2020, the average US-listed company had a cash conversion cycle of 78 days, though there was considerable variance by industry, as shown in Exhibit 5. Pharmaceutical manufacturers tend to have relatively long cash conversion cycles because they keep a considerable inventory of pharmaceuticals on hand. Airlines—a service business with minimal inventories of goods and mostly prepaid sales to customers in cash or credit cards—tend to have short cash conversion cycles.

Exhibit 5:

2020 S&P 1500 Industry Average Cash Conversion Cycles



▶ Expand / collapse Extended Description

Source: JPMorgan Working Capital Index 2021. Authors' analysis.

The ideal scenario is a short or even negative cash conversion cycle, which means that cash invested in inventory is quickly returned for subsequent investment. A negative cash conversion cycle can result from receiving cash from customers before—in some cases, *well* before—suppliers are paid. This scenario results in cash that can be used elsewhere, reducing the need for alternative financing to fund operations.

Inditex's Negative Cash Conversion Cycle

Industria de Diseño Textil, S.A., known as Inditex, is an apparel designer, manufacturer, and retailer listed on the stock exchanges of Madrid, Barcelona, Bilbao, and Valencia. Its largest brand concept is ZARA, which specializes in fast fashion.

Inditex has a unique business model in apparel, in which it leverages its vertically integrated supply chain to design, make, and sell numerous limited runs of inventories, enabling it to keep up with and influence customers' changing fashion tastes. Inditex sells off-the-rack items directly to customers (who pay with cash or credit cards), eschews stocking inventory to mitigate fashion risk, and sources raw materials from a vast number of suppliers over whom it has considerable leverage on payment terms. As a result, Inditex has a negative cash conversion cycle, as shown in the table below.

(in millions of EUR)	2021	2020
Accounts receivable	842	715
Inventories	3,042	2,321
Accounts payable	6,199	4,659
Sales	27,716	20,402
Cost of goods sold	11,902	9,013

Days of inventory on hand	93	94
Days payable outstanding	190	189
Cash conversion cycle	−86	−82

Issuers can shorten their cash conversion cycle in several ways, including the following:

- Reduce days of inventory on hand by discontinuing products or product lines with low or niche demand, by negotiating with suppliers to do more frequent deliveries in order to establish "just in time" inventory levels, and by using data analytics to improve customer demand forecasts and to rationalize stocking levels.
- Reduce days sales outstanding by offering prompt-payment discounts to customers, imposing late fees, tightening credit standards, imposing upfront deposits or accelerating installment payments, and working with third-party collection agencies.
- Increase days payable outstanding by negotiating supplier contracts for longer terms. This approach may be feasible by establishing preferred suppliers—purchasing more in volume in exchange for better terms. However, it may result in suppliers charging higher prices or asking for deposits. The ability to lengthen days payable outstanding is highly dependent on the power dynamics between the company and its suppliers. If a company can purchase a critical component from only one supplier who sells to many others, it may not have the ability to negotiate better terms.

While extending days payable outstanding can improve the cash conversion cycle, suppliers typically offer discounts for prompt payment, such as requiring payment in 30 days but offering a 2% discount if payment is received within 10 days. If a company forgoes this discount in favor of paying in 30 days, it is implicitly borrowing from the supplier for 30 − 10 = 20 days at the cost of the forgone discount. One strategy is to borrow from a third party (e.g., a bank) at a relatively low interest rate, pay the supplier early to receive the prompt-payment discount, and later repay the bank. Such a strategy avoids the high cost of supplier financing but preserves cash, as shown in the following example.

EXAMPLE 1

Keown Corporation—Internal versus External Financing

Keown Corporation is an established manufacturer of custom paddleboards operating in the North American market. Keown operates its own manufacturing plant in Canada and sells its paddleboards exclusively through its website to avoid the cost of retail locations.

Most of Keown's sales take place during the North American summer season from May to August. Keown's customers expect orders to be filled immediately, so it must maintain substantial inventory to start the summer season or risk losing sales to competitors. Given the seasonality of the business, Keown is particularly focused on meeting customers' needs.

Since Keown lacks the necessary cash to pay its suppliers within 10 days, the CFO must decide whether to borrow from its bank at an effective annual rate (EAR) of 7.7% to take the prompt-payment discount offered by its supplier of materials or pay in 30 days. The terms from the supplier are 2/10, net 30.

1. Should Keown use the bank loan and pay the supplier within 10 days to receive the 2% discount, or simply forgo the discount and pay the supplier in 30 days?

   Hide Solution

   Solution:

calculating the interest rate on a loan for which the interest cost is the forgone discount and the term is the additional time Keown gets to pay; in this case, 30 − 10 = 20 days.

$$\text{EAR of Supplier Financing} = \left(\left(1 + \frac{\text{Discount }\%}{100\% - \text{Discount }\%}\right)^{\frac{\text{Days in Year}}{\text{Payment Period}-\text{Discount Period}}}\right) - 1$$

$$\text{Effective Annual Rate of Supplier Financing} = \left(\left(1 + \frac{2\%}{100\% - 2\%}\right)^{\frac{365}{30-10}}\right) - 1$$

$$\text{Effective Annual Rate of Supplier Financing} = 0.446 \text{ or } 44.6\%$$

Since the effective annual rate of 44.6% on the supplier financing is significantly higher than the 7.7% interest rate on the bank loan, Keown should borrow from its bank. That way, it will still be able to preserve cash but will pay a far lower interest rate on the financing.

A long cash conversion cycle may reflect industry or business model characteristics, but a longer cycle relative to competitors and a lengthening over time are of particular concern for analysts. A longer cycle may signal worsening customer demand, deteriorating customer financial health or credit quality, or the loss of bargaining power with suppliers.

EXAMPLE 2

Cash Conversion for US Discount Retailers

Consider the following activity ratios for large US discount retailers Walmart Inc., Target Corporation, Costco Wholesale Corporation, The TJX Companies, and Ross Stores for the 2021 calendar year, as shown below.

	Walmart	Target	Costco	TJX	Ross
Days sales outstanding	5	2	3	7	2
Days of inventory on hand	48	68	29	63	61
Days payable outstanding	47	75	34	47	63

1. Which company has the shortest cash conversion cycle? The longest?

Hide Solution

Solution:

Cash conversion cycle = DSO + DOH − DPO

Cash conversion cycle for Walmart = 5 days + 48 days − 47 days = 6 days

Cash conversion cycle for Target = 2 days + 68 days −75 days = −5 days

Cash conversion cycle for Costco = 3 days + 29 days − 34 days = −2 days

Cash conversion cycle for TJX = 7 days + 63 days − 47 days = 23 days

Cash conversion cycle for Ross = 2 days + 61 days − 63 days = 0 days

Target has the shortest cash conversion cycle of the five companies.

2. These companies accept payment from customers at the point of sale in either cash or a debit or credit card. Does this fact align with the activity ratios presented?

Hide Solution

Solution:

immediately.

3. In examining annual reports, you find that approximately 54% of Costco's sales are food or food-related items, a larger percentage than that of the other companies, which sell a greater percentage of apparel and non-food items. Does this fact align with the activity ratios presented?

Hide Solution

Solution:

Yes. Companies like Costco that sell perishable goods tend to have a lower number of days of inventory on hand, because inventory must be sold soon after receiving it from suppliers or else it spoils.

In addition to the cash conversion cycle, another measure analysts use to assess the efficiency of business operations is the amount of **working capital** used by the firm, particularly its quantity relative to sales so that it can be compared across time and firms. There are several definitions of working capital in practice. We distinguish between **total working capital**, formally defined in Equation 2 as a broad measure, and **net working capital**, defined in Equation 3 as a measure that excludes items that are less related to the cash conversion cycle or business operations, such as cash, marketable securities, and short-term debt.

Current assets — 2
Minus: Current liabilities
Total Working Capital

Current assets, excluding cash and marketable securities — 3
Minus: Current liabilities, excluding short-term and current debt
Net Working Capital

To control for size and for comparability across firms, total or net working capital is often expressed as a percentage of annual sales.

Consider the information on Licht Vernieuwend N.V. for the fiscal year ending 31 December 20X2, as shown in Exhibit 6. The total working capital is the difference between the current assets of EUR335 million and the current liabilities of EUR220 million, or EUR115 million. The net working capital excludes cash, marketable securities, and short-term bank loans from the calculation, giving a net working capital of EUR225 million − EUR160 million = EUR65 million.

Exhibit 6:

Total and Net Working Capital Extracted from the Licht Vernieuwend N.V. Balance Sheet for the Year Ending 31 December 20X2

	(in millions of EUR)	
Cash	40	
Marketable securities	70	
Accounts receivable	85	85
Inventory	130	130
Prepaid accounts	10	10
Total	335	225
Accounts payable	130	130
Accrued expenses	30	30
Short-term bank loan	60	
Total	220	160

Net working capital	65

The cash conversion cycle and the ratio of working capital to sales are interrelated. Since receivables and inventories are often large components of short-term assets and payables are a large component of short-term liabilities, a short cash conversion cycle is associated with a low ratio of working capital to sales and vice versa.

Inditex's Negative Cash Conversion Cycle and Working Capital

Continuing the earlier example, owing to Inditex's business model—which includes prompt payment from customers for off-the-rack apparel, limited runs of inventories with fast turnover, and a vast number of suppliers over which it has leverage—Inditex has not only a negative cash conversion cycle but also negative net working capital and negative ratios of net working capital to sales, as shown in the table below.

(in millions of EUR)	2021	2020
Accounts receivable	842	715
Inventories	3,042	2,321
Accounts payable	6,199	4,659
Net working capital	−2,315	−1,623
As a % of sales	−8.4%	−8.0%
Sales	27,716	20,402
Cost of goods sold	11,902	9,013
Days sales outstanding	11	13
Days of inventory on hand	93	94
Days payable outstanding	190	189
Cash conversion cycle	−86	−82

A high ratio of working capital to sales may be a result of industry characteristics, such as in the spirits industry, where inventory must age for several years before being sold to customers, or in the pharmaceutical industry, where companies hold a large amount of inventory, sometimes to comply with regulations. Besides cases of necessity, as illustrated in Example 1, or cases of compliance with regulations, issuers are generally better off holding less working capital and either using capital elsewhere on higher-return projects or returning capital to investors.

Question Set

1. Classify each of the following actions by an issuer based on the likely effect on its cash conversion cycle:

	Effect on the Cash Conversion Cycle	
Action	Shorten	Lengthen
Offering larger discounts to its customers for payments received before the due date		
Paying suppliers sooner		
Lowering reliance on just-in-time inventory methods while increasing safety stocks of inventory		
Negotiating longer payment periods with its suppliers		

Action	Shorten	Lengthen
Tightening credit standards for its customers		

Hide Solution

Solution:

	Effect on the Cash Conversion Cycle	
Action	**Shorten**	**Lengthen**
Offering larger discounts to its customers for payments received before the due date	✓	
Paying suppliers sooner		✓
Lowering reliance on just-in-time inventory methods while increasing safety stocks of inventory		✓
Negotiating longer payment periods with its suppliers	✓	
Tightening credit standards for its customers	✓	

2. Identify the issuer with the longest cash conversion cycle and explain the effects of that length on that issuer relative to the other issuers.

	Issuer A	Issuer B	Issuer C
Days of inventory on hand	20	35	15
Days payable outstanding	15	10	5
Days sales outstanding	30	25	30

Hide Solution

Solution:

Using:

$$\text{Cash conversion cycle} = \text{Days of inventory on hand} + \text{Days sales outstanding} - \text{Days payable outstanding}$$

	Issuer A	Issuer B	Issuer C
Days of inventory on hand	20	35	15
Plus: Days sales outstanding	30	25	30
Minus: Days payable outstanding	15	10	5
Cash conversion cycle	35	50	40

Issuer B has the longest cash conversion cycle, 50 days. By having the longest cycle, Issuer B is more reliant on alternative financing to support its operations relative to the other issuers.

3. Assuming no change in days sales outstanding and days of inventory on hand, an issuer in need of cash flow that forgoes the discount offered by its vendor for payments within 10 days and chooses to pay on the due date in 30 days is:

A. shortening its cash conversion cycle.
B. lengthening its cash conversion cycle.
C. not affecting its cash conversion cycle.

Hide Solution

Solution:

B is correct. The issuer that uses the vendor financing by delaying payments is increasing its days payable outstanding and thus lengthening its cash

4. An issuer with limited cash flow is deciding which of its suppliers' credit terms are least costly. Which of the following credit terms offered to the issuer by its suppliers have the *lowest* effective interest rate?

A. 1/10, net 50
B. 2/15, net 40
C. 3/15, net 60

Hide Solution

Solution:

A is correct. The implicit financing cost that the issuer faces when forgoing a discount that the supplier or vendor offers is based on the amount of the forgone discount and length of the payment period beyond the discount period; 1/10, net 50 would permit the issuer to borrow for 50 − 10 = 40 days at a cost of only 1% of the purchase price (the forgone discount). The calculations for the cost of financing for each set of credit terms, expressed as an effective annual rate, are as follows:

$$\text{EAR of Supplier Financing} = \left(\left(1 + \frac{\text{Discount }\%}{100\% - \text{Discount }\%}\right)^{\frac{\text{Days in Year}}{\text{Payment Period–Discount Period}}}\right) - 1$$

$$\text{EAR of 1/10, net 50} = \left(\left(1 + \frac{1\%}{100\% - 1\%}\right)^{\frac{365}{50-10}}\right) - 1$$

$$\text{EAR of 1/10, net 50} = 0.096 \text{ or } 9.6\%$$

$$\text{EAR of 2/15, net 40} = \left(\left(1 + \frac{2\%}{100\% - 2\%}\right)^{\frac{365}{40-15}}\right) - 1$$

$$\text{EAR of 2/15, net 40} = 0.343 \text{ or } 34.3\%$$

$$\text{EAR of 3/15, net 60} = \left(\left(1 + \frac{3\%}{100\% - 3\%}\right)^{\frac{365}{60-15}}\right) - 1$$

$$\text{EAR of 3/15, net 60} = 0.280 \text{ or } 28.0\%$$

5. An issuer is comparing a bank loan at a rate of 15% with taking advantage of a supplier's terms of 1/14, net 30, paying on day 14. The best decision in terms of the lower cost of financing is to:

A. forgo the discount and use the supplier's financing to pay on day 30.
B. borrow from the bank to take advantage of the trade credit terms.
C. use either option because the cost of the bank loan and the cost of the trade credit are identical.

Hide Solution

Solution:

B is correct.

The cost of the supplier's trade credit, expressed as an effective annual rate, is 25.769%, which is higher than the 15% interest rate on the bank loan.

$$\text{EAR of Supplier Financing} = \left(\left(1 + \frac{\text{Discount }\%}{100\% - \text{Discount }\%}\right)^{\frac{\text{Days in Year}}{\text{Payment Period – Discount Period}}}\right) - 1$$

$$\text{EAR of 1/14, net 30} = \left(\left(1 + \frac{1\%}{100\% - 1\%}\right)^{\frac{365}{30-14}}\right) - 1$$

$$\text{EAR of 1/14, net 30} = 0.25769 \text{ or } 25.769\%$$

A and C are incorrect, because the bank loan rate of 15% is lower than the effective annual rate, 25.769%, on the supplier's trade credit.

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Cash	100
Marketable securities	20
Accounts receivable	600
Inventory	800
Prepaid expenses	30
Property, plant, and equipment	10,000
Intangibles	500
Total assets	12,050
Accounts payable	980
Accrued expenses	70
Short-term debt	1,000
Long-term debt	2,000
Shareholders' equity	8,000
Total liabilities and equity	12,050

7. The issuer's net working capital is *closest* to:

A. −500.
B. 380.
C. 500.

Hide Solution

Solution:

B is correct. Net working capital is defined as:

Current assets, excluding cash and marketable securities
Minus: Current liabilities, excluding short-term and current debt
Net Working Capital

Current assets, excluding cash and marketable securities:

Accounts receivable	600
Inventory	800
Prepaid expenses	30
Sum	1,430

Current liabilities, excluding short-term and current debt:

Accounts payable	980
Accrued expenses	70
Sum	1,050
Net working capital	380

A is incorrect, because cash, marketable securities, and short-term debt are mistakenly included in the calculation, resulting in the calculation of total working capital instead of net working capital.

C is incorrect, because cash and marketable securities are mistakenly included in the calculation.

Discuss Filter

Discussion

What do you want to discuss... (required)

It says the company “is increasing its DPO” (which is true).Then it says this “lengthens the cash conversion cycle”, which is false.
- Increasing DPO always reduces the CCC, because you're taking longer to pay your suppliers, effectively using their capital longer.

Created 5 months ago by Timothy Mayanja — 3 replies | Last Activity: 2 months ago
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AZ
Yes it should be. Relax as the explanation also explains A

Created 4 months ago by Abdulaziz Zahid — 0 replies | Last Activity: 4 months ago
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PE
3A

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RS
Correct Answer for Q3 should be A, agreed.

Created 5 months ago by Rishabh Sharma — 1 reply | Last Activity: 4 months ago
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HW
Answer to Q3 is A.
Link to CFA level 1 errata https://www.cfainstitute.org/programs/candidate-resources/submit-errata#230548828-1458461906
"A is correct. The issuer that uses the vendor financing by delaying payments is increasing its days payable outstanding and thus shortening its cash conversion cycle. The issuer is reducing its need for liquidity by taking advantage of the vendor financing at the cost of the forgone discount."

Created 6 months ago by Honghao Wu — 2 replies | Last Activity: 5 months ago
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NL
Question 3.. an issuer in need of cash flow + Chooses to pay on the due date in 30 days ==> 3A. shortening its cash conversion cycle.

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YL
why the answer to question 3 is B? no change in DOH and DSO, DPO increase, I think it is shortening its cash conversion cycle

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DH
I agreed that A is the correct answer.

Created 9 months ago by Danqing Hua — 1 reply | Last Activity: 6 months ago
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AC
Question set 3, answer should be A, not B.

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PD
is the answer to question set question number 3 correct? If DPO is increased, shouldn’t that shorten the cash conversion cycle

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